Creating Vertical Bar Diagram, Donut Chart & Editing Interactions in Power BI

1. Introduction

Power BI is a powerful business intelligence tool used to build interactive dashboards and visual reports. In this session, we focus on creating two specific visuals—**a Vertical Bar Diagram (Stacked Column Chart)** and **a Donut Chart**—based on the dataset provided in the case study. We also explore the process of **changing colors**, **editing interactions**, and **finalizing the dashboard** according to the project requirements.

The goal is to clearly understand how to:

1. Create a **vertical bar (column) chart** for *segment-wise regional distribution of profit*.
2. Create a **donut chart** for *regional distribution of orders*.
3. Apply **correct visual formatting rules** including color modifications, data labels, legend adjustments, chart titles, etc.
4. Understand and apply **Edit Interactions** so that slicers affect **only the KPIs** and do not change the charts or tables.
5. Finalize the report, fix style errors, and export the completed work.
6. Learn about embedding **Power BI reports in PowerPoint** for live interactive presentations.

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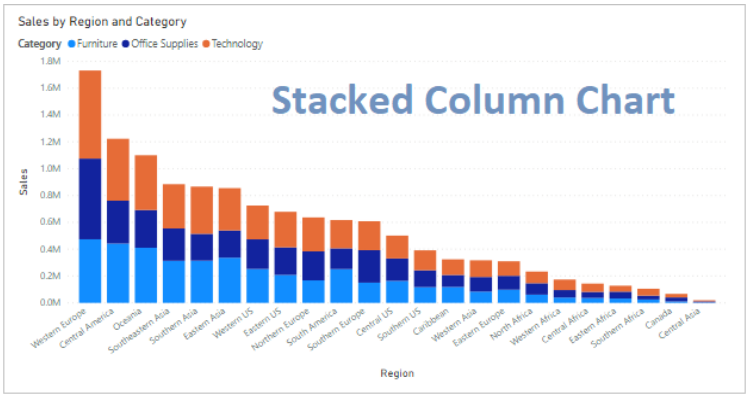
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2. Creating a Vertical Bar Diagram (Stacked Column Chart)

2.1 Requirement Understanding

Point number 7 in the case says:

> "Vertical bar diagram of *segment wise regional distribution of profit* without applying any filters."



A vertical bar diagram is simply a **Column Chart** in Power BI. Since the requirement mentions *segment-wise* and involves *regional distribution*, we must create a chart where:

- **X-axis** → Segment
- **Legend** → Region
- **Values** → Profit

Each segment will show bars split by region.

2.2 Creating the Vertical Bar Diagram

Step 1: Insert the Visual

1. Click on the **Stacked Column Chart** icon in the Visuals pane.
    - This is the second option under the bar/column chart category.
    - It is used for vertical bar diagrams and supports multiple categories within a bar.

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Step 2: Add Required Fields

1. **Drag Segment → X-axis**
    - Because we want *segment-wise* representation.
2. **Drag Profit → Values**
    - Profit is the quantitative measure that will form the height of the bars.
3. **Drag Region → Legend**
    - This distributes profit values across regions within each segment.

Now Power BI will automatically create stacked vertical bars.

2.3 Formatting the Column Chart

2.3.1 Modify the Chart Title

- Go to **Format → General → Title**.
- Change the title to:
  "Segment-wise Regional Distribution of Profit"
- Increase font size to **16 or 20** for visibility.

2.3.2 Applying Format Painter

You can copy formatting from an existing chart such as the line chart:

1. Select the line chart.
2. Click **Format Painter**.
3. Click on the column chart to apply the same formatting.
   This ensures consistency across visuals.

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3. Color Rules & Changing Legend Colors

3.1 Identifying the Problem

Initially, the colors in the column chart might look like this:

- Purple → West
- Orange → South
- Blue → East
- Light Blue → Central

These colors may match the colors used for **shipping modes** in the trend line chart.
This creates a data visualization issue because:

Thumb Rule of Data Visualization:

> **One color category should represent only one categorical variable.**
> **If the same colors are used for different variables, users may misinterpret the visuals.**

Thus, we must change the region color palette manually.

3.2 Changing the Colors Manually

Follow these steps:

1. Select the column chart.
2. Go to **Format → Columns → Colors → Show All**.
3. Under each region drop-down:
    - Select a unique color not used in other charts.
    - Repeat for:
        - Central
        - East
        - South
        - West

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This ensures *regional legend colors* are different from *shipping mode legend colors*.

3.3 Adding Data Labels

- Go to **Format → Data Labels**.
- Turn them **ON**.
- Optionally enable **Total Labels** as well.

Data labels help users easily read profit values directly from the chart.

3.4 Format the Y-Axis (Profit Axis)

- Go to **Format → Y-axis**.
- Change the title from "Sum of Profit" to simply:
  "Profit"
- Increase title font size for readability.

3.5 Resize the Visual

Resize the chart to free up space on the canvas for additional visuals such as the donut chart.

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4. Creating the Donut Chart (Regional Distribution of Orders)

4.1 Requirement Interpretation

Point number 8 says:

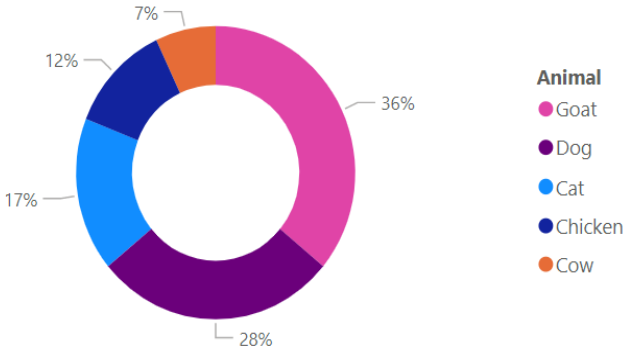
> "Donut chart of regional distribution of orders without any filters applied."

This means the donut chart must show how orders are split by region.

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4.2 Insert the Donut Chart

- Click on the **Donut Chart visual**.
- Place it in empty space on the report canvas.



4.3 Add Fields to the Donut Chart

We can measure orders in two ways:

1. **Order ID count**
    - Give the number of orders.

2. **Quantity (recommended)**
    - Represents order quantity, which is more meaningful.

You selected **Quantity**, so follow these steps:

1. Drag **Quantity → Values**.
2. Drag **Region → Legend**.

Power BI will create a donut segmented by region.

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4.4 Apply Format Painter

To maintain consistency:

- Select the column chart.
- Click **Format Painter**.
- Click on the donut chart.

This copies legend style, font sizes, border styles, and other formatting properties.

4.5 Edit Title

Change the chart title from “Sum of Quantity” to:

“Region-wise Quantity”

Adjust font size as needed.

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5. Edit Interactions in Power BI

5.1 Why Edit Interactions?

The requirement states:

> “Edit interactions must be controlled only for KPIs and no other charts and tables.”

Meaning:

- Slicers should affect **only the KPIs**.
- They should **NOT** change values in:
  - Column charts
  - Donut charts
  - Tables

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Thus, editing interactions helps freeze visuals so they don’t update with slicers.

5.2 How to Edit Interactions

Step-by-step Procedure

1. **Click on the slicer**
   (e.g., Year slicer)

2. Go to:
   Format → Edit Interactions

3. On all other visuals, two icons appear:

- **Circle with a line (None)** → disables interaction
- **Funnel icon (Filter)** → enables interaction

4. For the requirement:
    - Select **None** for:
        - Tables
        - Column chart
        - Donut chart
        - All other visuals except KPIs
5. Repeat this for **every slicer**:
    - Year
    - Region
    - State
    - Segment
    - Category
    - Subcategory

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This ensures only KPIs change values when slicers are used.

5.3 Validation

After disabling interactions:

- Change a slicer value.
- Check that:
    - KPIs update.
    - Column & donut charts **do not** update.

- Tables **do not** update.

This confirms correct configuration.

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6. Completing the Dashboard

6.1 Final Adjustments

- Resize visuals for aesthetic spacing.
- Verify color consistency.
- Fix any warnings or style alerts, such as:
  - “Style preset not found”
  - Formatting inconsistencies

Switching to the Default style preset usually resolves errors.

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6.2 Exporting the Dashboard

There are two export options:

1. **PDF Export**
   - Go to **File → Export → PDF**.
   - Exports a static snapshot of the report page.
2. **Interactive Report Sharing**
   - You must **Publish to Power BI Service**.
   - This requires signing in using a **Power BI Pro / Premium account**.

After publishing, you can collaborate with others through the Power BI service.

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7. Embedding Power BI Reports in PowerPoint

Microsoft allows embedding live Power BI visuals inside PowerPoint.

7.1 Purpose

- PowerPoint presentations are traditionally static.
- Embedded Power BI visuals allow:
    - Live filtering
    - Drill-down navigation
    - Page switching
    - Real-time updated visuals
      This makes presentations more interactive and engaging.

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7.2 Steps to Embed Power BI into PowerPoint

Step 1: Publish the Report

- Click **Publish** in Power BI Desktop.
- Sign in with your Pro account.
- Publish the file to Power BI Service.

Step 2: Open PowerPoint

- Prepare your presentation slides.

Step 3: Insert the Power BI Add-in

1. Go to **Insert → Get Add-ins**.
2. Search for **Power BI**.
3. Add the **Power BI add-in**.

This inserts a Power BI frame onto your slide.

Step 4: Sign-in

- Sign into PowerPoint using the *same* account used in Power BI.

Step 5: Embed the Report

- Choose the report from the workspace.
- Adjust the frame to fit the slide.

Step 6: Present Interactively

During the presentation you can:

- Click slicers
- Apply filters
- Navigate pages
- View tooltips

This makes storytelling more dynamic and accurate.

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7.3 Requirements & Limitations

- Requires **Power BI Pro or Premium license**.
- Viewers must have **access permissions**.
- Needs a stable **internet connection** to load visuals.
- Free users cannot embed live visuals.

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8. Creating Additional Report Pages

Power BI allows creation of multiple report pages for different analytical views.

Steps:

1. Click + **New Page** at the bottom.
2. Build new visuals, KPIs, and tables.
3. Only pin the important visuals to dashboards via the Power BI Service.

This helps organize large dashboards into multiple pages for each requirement.

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9. Conclusion

By following these steps, we successfully completed:

1. **Vertical Bar Diagram** for segment-wise regional distribution of profit.
2. **Donut Chart** for regional distribution of orders.
3. Application of correct **color rules** to avoid misinterpretation.
4. **Disabled slicer interactions** for all visuals except KPIs.
5. **Formatted titles, labels, legends, color themes**, and resolved style warnings.
6. Completed the **dashboard design** as per given requirements.
7. Learned how to:
    - Export reports as PDF
    - Publish to Power BI Service
    - Embed Power BI reports into PowerPoint for interactive storytelling

The end result is a completely functional, clean, well-formatted, and interactive Power BI dashboard that meets all the specifications of the case study.

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